

THILAGAN M.S.

Embedded Systems • Web Development • PCB Design

MY CONTACT

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EDUCATION

Bachelor of Engineering
Electrical & Electronics Engineering
Karpagam Institute of Technology
Expected: 2028
CGPA: 7.4 / 10

SKILLS

React.js / Next.js
Node.js / Express.js
MongoDB / PostgreSQL
HTML5 / CSS3 / JavaScript
Tailwind CSS / Bootstrap
AI Integration (APIs)
Git / GitHub / Vercel
Firebase / Supabase
Embedded Systems (IoT)
PCB Design & Simulation

TOOLS

Proteus
Altium Designer
MATLAB
KiCad
Web Development Stack
AI Tools for Automation

CERTIFICATIONS

Canva – Social Media & Product Design
Microsoft Advanced Excel
IBM Enterprise Design Thinking Practitioner
IBM Team Essentials for Designing AI Solutions

IoT – NPTEL

THILAGAN M.S.

Embedded Systems Developer • Web Developer • PCB Designer

Electrical & Electronics Engineering student with a multidisciplinary skill set spanning full-stack web development, embedded systems, and PCB design. Experienced in building scalable web applications with Node.js and React, designing and simulating circuits for IoT-based automation, and developing PCBs for real-world hardware products. Combines software and hardware fluency with AI-assisted development to deliver end-to-end, production-ready solutions.

CORE EXPERTISE

Frontend Development

React.js • Next.js • Tailwind CSS • Responsive Design • UI/UX Implementation • Framer Motion

Backend Development

Node.js • Express.js • REST APIs • Authentication • Database Design

Database & Cloud

MongoDB • PostgreSQL • Firebase • Supabase • Vercel • Netlify

AI Integration

OpenAI API • Google Gemini API • Claude API • Chatbot Integration • Prompt Engineering

Embedded Systems & IoT

Sensor Interfacing • Microcontroller Programming • IoT Automation • ML-based Monitoring • Circuit Simulation (Proteus, MATLAB)

PCB Design

Schematic Capture • PCB Layout & Routing • Altium Designer • KiCad • Prototyping & Testing

PROJECTS

Boiler Monitoring System - IoT & ML Automation

Real-time boiler parameter monitoring using IoT sensors with machine learning-based anomaly detection to improve safety and operational efficiency.

Resonant Wireless EV Charging System

Design and optimization of a resonant inductive wireless charging system for autonomous electric vehicles, incorporating solar integration.

RFID-Based Identification & Tracking System

RFID-based system for automated identification and access/asset tracking, integrating embedded hardware with a monitoring interface.

Solar-Powered Dewatering System for Mining Operations

Solar-powered automated dewatering solution designed to remove excess water from mining sites, reducing dependency on grid power.

KEY STRENGTHS

- ✓ Self-directed learner with passion for emerging technologies
- ✓ Cross-disciplinary expertise across software, embedded systems, and hardware
- ✓ Rapid prototyping and iterative development
- ✓ AI-assisted development and automation for smarter solutions
- ✓ Problem-solving mindset with focus on scalability
- ✓ User-centric design thinking across digital and physical products